ATTIA INAYAT

AI Automation Engineer | AI Systems Engineer | Agentic AI Engineer

+92-320-9176066 | attia.inayat23@gmail.com | linkedin.com/in/attia-inayat | github.com/AttiaInayat

Summary

AI Engineer and Machine Learning Developer specializing in Deep Learning, 3D Computer Vision reconstruction, Retrieval-Augmented Generation (RAG), and conversational AI systems. Experienced in training neural network architectures with PyTorch and TensorFlow, applying transfer learning pipelines, and developing end-to-end applications from spatial voxel generation to real-time voice automation. Solid foundation in Python, database integrations, and high-performance data processing libraries.

Skills

Languages: Python, JavaScript, C++, Java, SQL
AI, ML & Deep Learning: Machine Learning (ML), Deep Learning, Artificial Neural Networks (ANN), Natural Language Processing (NLP), Computer Vision, Transfer Learning, LangChain, LangGraph, OpenAI API, Google Gemini, Vector Databases (FAISS), Prompt Engineering
Voice AI & Automation: Retell AI, Vapi, Make.com, Airtable, Webhook Automations
Web & Frameworks: FastAPI, Node.js, Express.js, React
Databases & Caching: PostgreSQL, MongoDB, Redis, SQLAlchemy, Alembic
Data Science & ML Libraries: NumPy, Pandas, Matplotlib, Scikit-Learn, PyTorch, TensorFlow
Containerization: Docker, Docker Compose

Experience

Amazon Marketplace (USA) Remote
Virtual Assistant (Freelance) *Feb 2022 – Apr 2023*

- Managed wholesale business operations on the Amazon US marketplace, focusing on product sourcing, supplier relations, and day-to-day store operations.
- Scouted profitable, high-demand products and reached out directly to US brands and distributors to open wholesale accounts.
- Set up and fixed product listings, monitored inventory levels, and handled routine Seller Central maintenance to keep account health in good standing.

Projects

Blocksmith – AI 2D to 3D Minecraft Reconstruction (FYP) | *VQ-VAE, 3D Computer Vision, Python* 2026

- Built an automated data scraping and extraction pipeline to aggregate raw Minecraft builds, screenshots, and schematic assets from online repositories.
- Engineered, cleaned, and validated paired datasets aligning multi-perspective 2D structural renders with spatial 3D voxel representations.
- Trained a **VQ-VAE** (Vector Quantized Variational Autoencoder) architecture to reconstruct discrete 3D spatial voxel grids from limited 2D image inputs.
- Developed export pipelines translating predicted 3D voxel tensors directly into native **.schem** files for instant import into Minecraft worlds.

Context-Aware Conversational RAG System | *LangChain, FAISS, Python, FastAPI, Streamlit* 2025

- Built an end-to-end Retrieval-Augmented Generation (RAG) platform performing semantic vector search over unstructured document datasets.
- Engineered recursive document chunking, embeddings pipelines, and vector similarity retrieval via **FAISS** to eliminate hallucinations.
- Constructed an interactive query interface using Streamlit backed by asynchronous inference routing in **FastAPI**.

Autonomous Restaurant Voice AI & Order System | *Retell AI, Vapi, Make.com, Airtable, Webhooks* 2026

- Constructed a real-time conversational voice agent automating restaurant telephone inquiries, menu recommendations, and order placements.
- Configured custom prompt schemas and function calling in **Vapi** and **Retell AI** to extract caller preferences, order items, and delivery addresses.
- Automated post-call reservation logging and customer record syncs directly into **Airtable** via instant webhook triggers.

Education

COMSATS University Islamabad, Lahore Campus Lahore, Pakistan
*Bachelor of Science in Artificial Intelligence – **CGPA: 3.66*** *2023 – 2026*